

Miguel Rivera-Lanas

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EDUCATION

Carnegie Mellon University - Pittsburgh, PA

GPA: 3.8

Master of Science: Data Science and Public Policy

May 2025 (Expected)

- **Relevant Coursework:** Machine Learning: Intro to Machine Learning (ML Dept.), Intro to Deep Learning (ML Dept.)
Public Policy: Policies of Wireless Systems, Federal Budget Policy, Management: Optimization, Decision & Risk Modeling
- **Teaching Assistant Positions** → Summer 2024 - Math & CS for Machine Learning (ML Dept.), Fall 2024 - Privacy in the Digital Age (Information Systems), Spring 2025: - Engineering Privacy in Software (CS Dept.)

University of Pennsylvania - Philadelphia, PA

GPA: 3.5

Bachelor of Arts: Economics, Minor in Statistics, International Relations

May 2017

- **Relevant Coursework** → Economics: Microeconomics, Macroeconomics, Econometrics, Modern Regression, Math: Calculus I/II, Probability, Political Science: Modern Political Thought, Latin America since 1791

PROFESSIONAL EXPERIENCE

Allegheny County Dept. Health & Human Services (DHS)

Pittsburgh, PA

Data Science Intern - Office of Technology & Analytics

November 2024 - Present

- Developed ML models to identify top K individuals most at risk of facing homelessness in Allegheny County.
- Improved Random Forest precision at K=400 by 10%+ via feature and hyperparameter engineering.
- Developed fairness assessments (FPR and predictive parity) across demographic and custom categories.
- Designed ETL pipeline for model retraining and deployment through shell-based procedures in Bash.

Carnegie Mellon University

Pittsburgh, PA

Research Assistant – Privacy Engineering

Jan. 2024 – Sept 2024

- Developed privacy threat framework focused on usability of web, mobile, and IoT privacy control interfaces.
- Presented at 2024 Symposium on Usable Privacy, the leading usable privacy conference. See paper [here](#).

UPenn Computational Social Science Lab

Philadelphia, PA

Research Data Scientist, Data Engineer

April 2022 – Aug 2023

- Developed real time ETL pipeline for >50 TB of unstructured TV transcripts using Apache Spark, resulting in over 50% monthly savings in AWS EC2 compute and S3 storage costs.
- Developed Selenium web-crawler to scrape YouTube videos - scaled to review over 13K hours of content.
- Evaluated interpretability of hierarchical clustering of online political content using NLP-based embeddings.
- Published research on online political radicalization at leading social science conference. See paper [here](#).

Point72 Asset Management

New York City, NY

Portfolio Data Scientist

Sept 2019 - April 2022

- Built mixed-effect regression models to forecast monthly crude oil production across the Texas Permian Basin.
- Developed Random Forest models to forecast same-store sales and other KPIs to inform investment decisions.
- Delivered real-time analytics with event-driven ETL pipelines using Airflow, Jenkins, AWS, and Databricks.

AllianceBernstein Asset Management

New York City, NY

Investment Research Associate, Fixed Income

June 2017 - August 2019

- Evaluated reliability of bond pricing mixed-effect regression model within niche market segments.
- Developed real-time corporate bond supply and demand metrics from unstructured order book database.

PROJECTS

Volunteer Developer — Statisticians without Borders

Nov 2020 - March 2021

- Developed an AWS dashboard enabling Indian health organizations to publish COVID-19 insights – article [here](#).

Intro to Deep Learning (CMU) – Fairness in Federated Deep Learning

Aug 2024 - Dec 2024

- Developed a novel federated learning approach for training neural networks using Flower AI - balancing accuracy with fairness measures in crime recidivism prediction context.

SKILLS

Scripting: Python, R, Scala, Bash **DevOps:** Git, Airflow, Docker, Jenkins **Cloud:** AWS (S3, EC2, Lambda, SNS, SES, Glue, Athena) **Database:** PostgreSQL, MongoDB **Machine Learning:** Sklearn **Deep Learning:** PyTorch. **Web Development:** Flask, Plotly/Dash **Web Scraping:** Selenium